

Response to NHSJS Reviewer Feedback

Dear Editor and Reviewers,

Thank you for the detailed review. I treated the decision letter as a roadmap for a full rebuild rather than a light revision. The manuscript now uses correct bacterial terminology, explicit RefSeq accessions, a larger complete-genome dataset, a deterministic and documented sonification map, synthetic mutation ground truth, shuffled negative controls, and quantitative statistics. The claims have been narrowed to what the data support.

Major Changes

- Replaced the original three-record comparison with 20 complete RefSeq *Xylella fastidiosa* genomes.
- Removed unsupported viral terminology and corrected the organism biology throughout.
- Replaced qualitative sound descriptions with pairwise feature distances, synthetic mutation-rate prediction, and shuffled null controls.
- Removed the incorrect claim that BLASTp validates audio output.
- Added a transparent codon-to-MIDI mapping, preprocessing rules, scripts, metadata files, and reproducible output tables.
- Expanded and modernized the reference list, including the sonification papers requested by the reviewer.

Point-by-Point Response

1. Reviewer concern: Aim and hypothesis unclear. Response: The Introduction now states a narrow hypothesis: audio-feature distance should increase with sequence divergence and distinguish synthetic mutations from shuffled null controls.
2. Reviewer concern: Organism described incorrectly. Response: The manuscript now identifies *Xylella fastidiosa* as a Gram-negative, xylem-limited bacterium and removes viral language.
3. Reviewer concern: Abstract too vague. Response: The abstract is now structured with context, objective, methods, results, and conclusion, including sample size and numerical findings.
4. Reviewer concern: Unsupported disease/accession claims. Response: Disease claims are tied to the *Xylella* literature, and Table 1 lists all accessions, strains, hosts or sources, locations, and sequence lengths.
5. Reviewer concern: Sample size inadequate. Response: The analysis now uses 20 complete RefSeq genomes plus 300 synthetic mutation records and shuffled null controls.
6. Reviewer concern: Incomplete records treated as genomes. Response: The selected dataset is restricted to RefSeq assemblies labeled Complete Genome. The problematic short accessions from the original draft are no longer used as genomes.

7. Reviewer concern: Reference genome undefined. Response: The revised design does not use an undefined 'minimally mutated' reference. Each genome serves as its own baseline for synthetic mutation testing.
8. Reviewer concern: Missing sonification literature. Response: Temple 2017, Martin 2021, and Plaisier 2021 were added and discussed.
9. Reviewer concern: Auditory value not justified. Response: The Discussion now frames sonification as an exploratory and educational display layer, not a replacement for alignment or variant calling.
10. Reviewer concern: Preprocessing not reproducible. Response: The Methods now specify FASTA parsing, base filtering, standardized 180,000-base segments, software files, output paths, and random seed.
11. Reviewer concern: Mapping not specified. Response: The Methods now define the codon ordering, MIDI pitch formula, features, and preview MIDI generation.
12. Reviewer concern: Transcription rationale unclear. Response: The revised pipeline no longer claims biological transcription to mRNA; it analyzes DNA codons directly.
13. Reviewer concern: BLASTp validation inappropriate. Response: This claim was removed. Validation now uses known synthetic mutation rates and negative controls.
14. Reviewer concern: No negative controls. Response: Shuffled-sequence null controls preserving nucleotide composition were added.
15. Reviewer concern: No quantitative acoustic features. Response: Pitch distribution, interval statistics, entropy, and pitch histograms are now measured.
16. Reviewer concern: No statistical tests. Response: The Results now report pairwise correlation, holdout R2, mean absolute error, and median distances for mutation and null conditions.
17. Reviewer concern: Overclaims. Response: Claims about pathogenicity, virulence detection, and diagnostic use were removed or explicitly limited.
18. Reviewer concern: Reference list too short. Response: The manuscript now contains 22 references, including primary *Xylella* genome/pathology sources and sonification/auditory-display sources.

Sincerely,

Albert Zhang